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Sanitary washing device

Abstract

A sanitary washing device includes a nozzle, a heat exchanger, a cold water flow channel, a switching element, a control board, a heat transfer plate, and a casing. The heat exchanger uses alternating current power to heat the wash water supplied to the nozzle. The cold water flow channel supplies the wash water to the heat exchanger from a water supply source. The switching element switches a supply state of the alternating current power to the heat exchanger. A controller is mounted to the control board. The controller is operated by direct current power. The controller controls the switching element. The heat transfer plate is configured to dissipate heat of the switching element. The switching element is mounted to the control board. The cold water flow channel includes a cooling part. The cooling part water-cools the heat transfer plate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2022-097336, filed on Jun. 16, 2022; the entire contents of which are incorporated herein by reference.

FIELD

(2) Embodiments described herein relate generally to a sanitary washing device.

BACKGROUND

(3) In a known sanitary washing device that includes a nozzle discharging wash water for a private part wash, a heat exchanger uses AC power to heat the wash water. In such a sanitary washing device, for example, a switching element such as a triac or the like switches the supply state of the AC power to the heat exchanger. Normally, such a switching element is controlled by a controller that is operated, similarly to the functional units included in the sanitary washing device, by DC power.

(4) In sanitary washing devices of recent years, it is desirable to make the casing smaller to improve the designability. To make the casing smaller while storing the various functional units within, it may be considered to mount the switching element and the controller, which are conventionally mounted on separate control boards, on one control board. However, the switching control by the switching element generates heat; and the heat from the switching element may affect the controller if the switching element and the controller are mounted on one control board.

SUMMARY

(5) According to the embodiment, a sanitary washing device includes a nozzle, a heat exchanger, a cold water flow channel, a switching element, a control board, a heat transfer plate, and a casing. The nozzle discharges wash water for a private part wash. The heat exchanger uses alternating current power to heat the wash water supplied to the nozzle. The cold water flow channel supplies the wash water to the heat exchanger from a water supply source. The switching element switches a supply state of the alternating current power to the heat exchanger. A controller is mounted to the control board. The controller is operated by direct current power. The controller controls the switching element. The heat transfer plate is configured to dissipate heat of the switching element. The casing stores the nozzle, the heat exchanger, the cold water flow channel, the switching element, the control board, and the heat transfer plate. The switching element is mounted to the control board. The cold water flow channel includes a cooling part. The cooling part water-cools the heat transfer plate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a perspective view illustrating a toilet device that includes a sanitary washing device according to an embodiment;

(2) FIG. **2** is a block diagram illustrating relevant components of the sanitary washing device according to the embodiment;

(3) FIG. **3** is a perspective view illustrating a portion of the sanitary washing device according to the embodiment;

(4) FIG. **4** is a plan view illustrating a portion of the sanitary washing device according to the embodiment;

(5) FIG. **5** is a perspective view illustrating the periphery of the switching element of the sanitary washing device according to the embodiment;

(6) FIG. **6** is a plan view illustrating the periphery of the switching element of the sanitary washing device according to the embodiment;
(7) FIG. **7** is a cross-sectional view illustrating a portion of the sanitary washing device according to the embodiment;
(8) FIG. **8** is a cross-sectional view illustrating a portion of the sanitary washing device according to the embodiment;
(9) FIG. **9** is a perspective view illustrating a portion of the cold water flow channel of the sanitary washing device according to the embodiment; and
(10) FIG. **10** is a perspective view illustrating a portion of the casing of the sanitary washing device according to the embodiment.
DETAILED DESCRIPTION
(11) A first invention is a sanitary washing device, including a nozzle discharging wash water for a private part wash, a heat exchanger using AC power to heat the wash water supplied to the nozzle, a cold water flow channel supplying the wash water to the heat exchanger from a water supply source, a switching element switching a supply state of the AC power to the heat exchanger, a control board to which a controller is mounted, a heat transfer plate configured to dissipate heat of the switching element, and a casing storing the nozzle, the heat exchanger, the cold water flow channel, the switching element, the control board, and the heat transfer plate, wherein the controller is operated by DC power and controls the switching element, the switching element is mounted to the control board, and the cold water flow channel includes a cooling part that water-cools the heat transfer plate.
(12) According to the sanitary washing device, by mounting the switching element to the control board to which the controller is mounted, the number of components can be reduced and the casing can be made smaller than when the switching element and the controller are mounted to separate control boards. Also, the heat of the switching element can be effectively dissipated by providing a heat transfer plate for dissipating the heat of the switching element and by providing a cooling part that water-cools the heat transfer plate. Accordingly, the effects of the heat from the switching element on the controller can be suppressed even when the switching element and the controller are mounted to one control board. Because the cold water flow channel includes a cooling part, the heat transfer plate can be water-cooled by utilizing wash water supplied from the water supply source to the heat exchanger before being heated. Accordingly, it is unnecessary to separately provide a part to cool the heat transfer plate; and the casing can be made smaller. The heat transfer plate (the switching element) can be cooled more efficiently when the heat transfer plate is water-cooled than when the heat transfer plate is air-cooled.
(13) A second invention is the sanitary washing device of the first invention, wherein the heat exchanger is an instantaneous heat exchanger, and a distance between the heat exchanger and the switching element is greater than a distance between the cooling part and the switching element.
(14) According to the sanitary washing device, by setting the distance between the heat exchanger and the switching element to be greater than the distance between the cooling part and the switching element, the heat exchanger, i.e., a heat source, can be located distant to the switching element, and the cooling part can be located proximate to the switching element. The effects of the heat from the heat exchanger on the controller mounted to the same control board as the switching element can be suppressed thereby. Also, by not positioning the heat exchanger between the cooling part and the switching element, the heating of the water flowing through the cooling part by the heat from the heat exchanger can be suppressed. The heat transfer plate (the switching element) can be more efficiently cooled thereby.
(15) A third invention is the sanitary washing device of the first or second invention, wherein the casing includes a substrate storage part configured to store the control board in an upright state, the substrate storage part is located at a back part of the casing, and the switching element is mounted to a lower part of the control board.

(16) According to the sanitary washing device, for example, when the height of the back of the casing is greater than the height of the front of the casing, the casing can be prevented from becoming larger by providing the substrate storage part that can store the control board in an upright state at the back part of the casing; and the space inside the casing can be effectively used. Because it is common for the cold water flow channel to be located at the lower part of the casing, the cooling part can be made smaller by mounting the switching element to the lower part of the control board in the upright state compared to when the switching element is mounted to the upper part of the control board in the upright state. The casing can be made smaller thereby.
(17) A fourth invention is the sanitary washing device of the third invention, further including a valve unit located in the cold water flow channel, wherein the heat transfer plate is located at a position at which the heat transfer plate overlaps the valve unit in a longitudinal direction, and the switching element is located at a position at which the switching element overlaps the heat transfer plate in the longitudinal direction.
(18) According to the sanitary washing device, by providing the heat transfer plate at a position at which the heat transfer plate overlaps the valve unit in the longitudinal direction and by providing the switching element at a position at which the switching element overlaps the heat transfer plate in the longitudinal direction, the cooling part can be made smaller than when the heat transfer plate is located at a position at which the heat transfer plate does not overlap the valve unit in the longitudinal direction or when the switching element is located at a position at which the switching element does not overlap the heat transfer plate in the longitudinal direction. The casing can be made smaller thereby.
(19) A fifth invention is the sanitary washing device of the third invention, wherein the casing includes a water receiving part, at least a portion of the water receiving part is positioned below the heat transfer plate, the water receiving part is configured to receive condensation water generated at the heat transfer plate, and the heat transfer plate is inclined downward from the switching element toward the water receiving part.
(20) According to the sanitary washing device, the flow of the condensation water generated at the heat transfer plate toward the control board side can be suppressed by providing the water receiving part, which is configured to receive the condensation water generated at the heat transfer plate, below the heat transfer plate, and by setting the heat transfer plate to be inclined downward from the switching element toward the water receiving part. Malfunction of the control board due to condensation water can be suppressed thereby.
(21) A sixth invention is the sanitary washing device of the third invention, further including a toilet lid, and a toilet lid opening/closing part configured to open and close the toilet lid, wherein the control board is located at one side of the casing in a lateral direction, and the toilet lid opening/closing part is located at another side of the casing in the lateral direction.
(22) According to the sanitary washing device, the space inside the casing can be effectively used by providing the control board at one side of the casing in the lateral direction and by providing the toilet lid opening/closing part at the other side of the casing in the lateral direction. The casing can be made smaller thereby.
(23) Hereinafter, embodiments of the invention will be described with reference to the drawings. It is noted that, in each figure, similar components are denoted by the same reference numerals, and detailed description thereof will be omitted as appropriate.
(24) FIG. **1** is a perspective view illustrating a toilet device that includes a sanitary washing device according to an embodiment.
(25) As illustrated in FIG. **1**, the toilet device **500** according to the embodiment includes the sanitary washing device **100** and a western-style sit-down toilet (for convenience of description hereinbelow, called, simply the “toilet”) **200**. The sanitary washing device **100** is located on the toilet **200**. The sanitary washing device **100** includes a casing **10**, a toilet seat **20**, and a toilet lid **25**.
(26) In this specification, “up”, “down”, “front”, “back”, “right”, and “left” are directions when

viewed by a user sitting on the toilet seat **20** as illustrated in FIG. **1**.
(27) The casing **10** includes a case plate **11** and a case cover **12**. The case plate **11** is positioned at the lower part of the casing **10**. The case plate **11** is placed on the back part of the toilet **200**. The case cover **12** is located on the case plate **11** and covers the case plate **11** from above. The space of the casing **10** surrounded with the case plate **11** and the case cover **12** stores functional units such as a nozzle **30** described below, etc.
(28) The toilet seat **20** is pivotally supported to be openable and closable with respect to the casing **10** via a toilet seat opening/closing part **21** described below. The toilet lid **25** is pivotally supported to be openable and closable with respect to the casing **10** via a toilet lid opening/closing part **26** described below. FIG. **1** illustrates a state in which the toilet seat **20** and the toilet lid **25** are closed. The toilet lid **25** covers the toilet seat **20** from above in the state in which the toilet seat **20** and the toilet lid **25** are closed.
(29) FIG. **2** is a block diagram illustrating relevant components of the sanitary washing device according to the embodiment.
(30) In FIG. **2**, the relevant components of the water channel system and the relevant components of the electrical system of the sanitary washing device **100** are illustrated together.
(31) As illustrated in FIG. **2**, the sanitary washing device **100** includes the nozzle **30**, a heat exchanger **40**, a cold water flow channel **50**, a warm water flow channel **55**, a switching element **60**, and a control board **70**. The nozzle **30**, the heat exchanger **40**, the cold water flow channel **50**, the warm water flow channel **55**, the switching element **60**, the control board **70**, etc., are stored inside the casing **10**.
(32) The nozzle **30** discharges wash water for a private part wash. A discharge port **31** for discharging the wash water is provided in the tip portion of the nozzle **30**. The nozzle **30** discharges, through the discharge port **31**, the wash water supplied from a water supply source WS such as a service water line, a water storage tank, or the like to wash a private part (e.g., the “bottom” or the like) of a user sitting on the toilet seat **20**.
(33) A nozzle wash chamber **32** is located at the periphery of the nozzle **30**. The nozzle wash chamber **32** can sterilize or wash the outer circumferential surface (the body) of the nozzle **30** by discharging sterilizing water or wash water from a not-illustrated water discharger provided inside the nozzle wash chamber **32**. The nozzle wash chamber **32** can sterilize or wash the discharge port **31** part of the nozzle **30** in a state in which the nozzle **30** is stored inside the casing **10**.
(34) The nozzle **30** advances and retracts due to the drive force of a nozzle motor **33**. The drive force of the nozzle motor **33** causes the nozzle **30** to advance into the bowl of the toilet **200** or causes the nozzle **30** to retreat into the casing **10**. The nozzle **30** is stored inside the casing **10** in the retracted state.
(35) The heat exchanger **40** uses AC power to heat the wash water supplied to the nozzle **30**. The heat exchanger **40** is, for example, a ceramic heater. The heat exchanger **40** is, for example, an instantaneous heat exchanger that does not include a tank. An instantaneous heat exchanger heats the water flowing through the interior without filling a tank or the like with water. The heat exchanger **40** may be, for example, a hot water storage-type heat exchanger that includes a tank. The hot water storage-type heat exchanger heats water stored in a tank.
(36) In the example, the heat exchanger **40** includes a first heater **41** and a second heater **42**. The first heater **41** is, for example, a main heater having a large output. The second heater **42** is, for example, a sub-heater having a small output. The resistance value of the second heater **42** is, for example, less than the resistance value of the first heater **41**. The heat exchanger **40** uses the first heater **41** and the second heater **42** to heat the wash water supplied from the water supply source WS. The number of heaters provided in the heat exchanger **40** is not limited to two and may be one, three, or more.
(37) The cold water flow channel **50** supplies the wash water from the water supply source WS to the heat exchanger **40**. The cold water flow channel **50** is located between the water supply source

WS and the heat exchanger **40** and connects the water supply source WS and the heat exchanger **40**. Wash water (cold water) before being heated by the heat exchanger **40** flows in the cold water flow channel **50**. The cold water flow channel **50** is a flow channel at the upstream side of the heat exchanger **40**.

(38) A valve unit **51** is located in the cold water flow channel **50**. The valve unit **51** is an electrically controllable valve. The valve unit **51** opens and closes to control the start and stop of the supply of the wash water from the water supply source WS to the heat exchanger **40**. The valve unit **51** includes, for example, an electromagnetic valve. The valve unit **51** may include, for example, an electromagnetic valve and a pressure reducing valve.

(39) The warm water flow channel **55** supplies the wash water from the heat exchanger **40** to the nozzle **30**. The warm water flow channel **55** is located between the heat exchanger **40** and the nozzle **30** and connects the heat exchanger **40** and the nozzle **30**. The wash water (the warm water) after being heated by the heat exchanger **40** flows in the warm water flow channel **55**. The warm water flow channel **55** is a flow channel at the downstream side of the heat exchanger **40**.

(40) A flow rate switch valve **56** that regulates the flow rate and a flow channel switch valve **57** that starts and stops the water supply to the nozzle **30** and the nozzle wash chamber **32** and switches the water supply destination are located in the warm water flow channel **55**. The flow rate switch valve **56** regulates the flow rate of the water flowing toward the nozzle **30**. The flow channel switch valve **57** can switch the water supply destination (the connection destination of the warm water flow channel **55**) to one of the nozzle **30** or the nozzle wash chamber **32**. The flow rate switch valve **56** and the flow channel switch valve **57** may be provided as one unit.

(41) The switching element **60** switches the supply state of the AC power to the heat exchanger **40**. More specifically, the switching element **60** switches the supply state of the AC power to the heater of the heat exchanger **40**. The switching element **60** switches between a state in which AC power is supplied to the heater of the heat exchanger **40**, and a state in which AC power is not supplied to the heater of the heat exchanger **40** (i.e., a state in which the supply of the AC power to the heater is stopped). The switching element **60** thereby controls the start and stop of the heating of the heat exchanger **40**.

(42) The switching element **60** is, for example, a semiconductor switch. The switching element **60** includes, for example, a thyristor or a triac. The switching element **60** may be, for example, any element that can control the on/off of the current and can allow the current to flow in at least one direction. For example, the switching element **60** may be configured by combining multiple semiconductor switches, etc.

(43) In the example, the switching element **60** includes a first switching element **61** and a second switching element **62**. The first switching element **61** is electrically connected with the first heater **41** and switches the supply state of the AC power to the first heater **41**. The second switching element **62** is electrically connected with the second heater **42** and switches the supply state of the AC power to the second heater **42**. It is sufficient for the number of the switching elements **60** to be equal to the number of heaters included in the heat exchanger **40**; the number of the switching elements **60** is not limited to two and may be one, three, or more.

(44) The sanitary washing device **100** includes the toilet seat opening/closing part **21** and the toilet lid opening/closing part **26**. The toilet seat opening/closing part **21** includes, for example, an electric opening/closing unit for electrically opening and closing the toilet seat **20**. The toilet seat opening/closing part **21** may include, for example, a damper mechanism for slowly opening and closing the toilet seat **20**. The toilet lid opening/closing part **26** includes, for example, an electric opening/closing unit for electrically opening and closing the toilet lid **25**. The toilet lid opening/closing part **26** may include, for example, a damper mechanism for slowly opening and closing the toilet lid **25**. The toilet seat opening/closing part **21** and the toilet lid opening/closing part **26** are provided as necessary and are omissible.

(45) A controller **71** for controlling the components of the sanitary washing device **100** described

above is mounted to the control board **70**. The controller **71** is operated by DC power. The controller **71** is electrically connected with the switching element **60** (the first switching element **61** and the second switching element **62**) and controls the switching element **60** (the first switching element **61** and the second switching element **62**). For example, the controller **71** is electrically connected with the valve unit **51** and controls the valve unit **51**. For example, the controller **71** is electrically connected with the flow rate switch valve **56** and the flow channel switch valve **57** and controls the flow rate switch valve **56** and the flow channel switch valve **57**. For example, the controller **71** is electrically connected with the nozzle motor **33** and controls the nozzle motor **33**. For example, the controller **71** is electrically connected with the toilet seat opening/closing part **21** and the toilet lid opening/closing part **26** and controls the toilet seat opening/closing part **21** and the toilet lid opening/closing part **26**.

(46) For example, by operating an operation part **75**, the user can start and stop the discharge of the wash water from the nozzle **30**. For example, by operating the operation part **75**, the user can set the temperature of the wash water heated in the heat exchanger **40**. The operation part **75** is, for example, a remote control used by being mounted to a wall of a toilet room, etc. The operation part **75** may be, for example, an operation panel formed to have a continuous body with the casing **10** of the sanitary washing device **100**, etc.

(47) AC power from a power supply PS is supplied to the heat exchanger **40**. The power supply PS is an AC power supply. The power supply PS is, for example, a commercial power source of AC 100 V (effective value).

(48) The AC power from the power supply PS is supplied to the heat exchanger **40** via an input circuit **72**. The input circuit **72** is located on the path between the power supply PS and the heater of the heat exchanger **40**. The heater (the first heater **41** and the second heater **42**) of the heat exchanger **40** is connected to the input circuit **72** via the switching element **60** (the first switching element **61** and the second switching element **62**).

(49) A portion of the AC power supplied from the power supply PS to the input circuit **72** is supplied to the heater (the first heater **41** and the second heater **42**) of the heat exchanger **40** via the switching element **60** (the first switching element **61** and the second switching element **62**). The AC power that is supplied from the power supply PS (the input circuit **72**) is supplied by the first switching element **61** to the first heater **41** in the on-state. In the off-state, the first switching element **61** stops the supply to the first heater **41** of the AC power supplied from the power supply PS (the input circuit **72**). The AC power that is supplied from the power supply PS (the input circuit **72**) is supplied by the second switching element **62** to the second heater **42** in the on-state. In the off-state, the second switching element **62** stops the supply to the second heater **42** of the AC power supplied from the power supply PS (the input circuit **72**).

(50) Another portion of the AC power supplied from the power supply PS to the input circuit **72** is converted into DC power by a power supply circuit **73** and supplied to the controller **71**. The power supply circuit **73** is located on the path between the input circuit **72** and the controller **71**, converts the AC power supplied from the power supply PS into DC power, and supplies the DC power after the conversion to the controller **71**. For example, the controller **71** operates based on the DC power supplied from the power supply circuit **73**.

(51) The input circuit **72** and the power supply circuit **73** are located at the control board **70**. The input circuit **72** and the power supply circuit **73** supply a portion of the AC power supplied from the power supply PS to the heat exchanger **40** as AC power, and supply another portion of the AC power supplied from the power supply PS to the controller **71** as DC power.

(52) The controller **71** may operate based on DC power supplied from a DC power supply (e.g., a battery or the like) other than the power supply PS. In such a case, the power supply circuit **73** is omissible.

(53) According to the embodiment, the switching element **60** is mounted to the control board **70**. In other words, the switching element **60** and the controller **71** are mounted to the same one control

board **70**. Thus, by mounting the switching element **60** to the control board **70** to which the controller **71** is mounted, the number of components can be reduced and the casing **10** can be made smaller than when the switching element **60** and the controller **71** are mounted to separate control boards.

(54) On the other hand, the switching element **60** generates heat due to the control of switching the supply state of the AC power to the heat exchanger **40**. Therefore, there is a risk that the controller **71** may be affected by the heat from the switching element **60** when the switching element **60** and the controller are mounted to one control board **70**. Therefore, according to the embodiment, a cooling mechanism for cooling the switching element **60** is provided.

(55) The cooling mechanism for cooling the switching element **60** will now be described.

(56) FIG. **3** is a perspective view illustrating a portion of the sanitary washing device according to the embodiment.

(57) FIG. **4** is a plan view illustrating a portion of the sanitary washing device according to the embodiment.

(58) FIG. **5** is a perspective view illustrating the periphery of the switching element of the sanitary washing device according to the embodiment.

(59) FIGS. **3** and **4** illustrate the state in which the case cover **12** is removed.

(60) As illustrated in FIGS. **3** to **5**, a heat transfer plate **80** for dissipating the heat of the switching element **60** is located below the switching element **60**.

(61) The heat transfer plate **80** is, for example, a plate made of metal. In the example, the heat transfer plate **80** is fixed with respect to the switching element **60** and contacts the switching element **60**. It is sufficient for the heat transfer plate **80** to be located in the vicinity of the switching element **60** (e.g., a position to which the heat of the switching element **60** is conducted); the heat transfer plate **80** may not contact the switching element **60**. The heat transfer plate **80** may not be fixed with respect to the switching element **60**.

(62) The heat transfer plate **80** includes a first part **81** and a second part **82**. For example, the first part **81** extends substantially horizontally frontward from the control board **70**. For example, the first part **81** is positioned below the switching element **60**. The first part **81** is fixed with respect to the switching element **60** and contacts the switching element **60**. For example, the first part **81** is fixed to the switching element **60** from below by screwing, etc. The second part **82** extends to be inclined downward and frontward from the first part **81**.

(63) The cold water flow channel **50** includes a cooling part **52** that water-cools the heat transfer plate **80**. The cooling part **52** is fixed with respect to the second part **82** of the heat transfer plate **80** and contacts the second part **82** of the heat transfer plate **80**. In the example, the cooling part **52** is located between the valve unit **51** and the heat exchanger **40**. That is, in the example, the cooling part **52** is located downstream of the valve unit **51**. The cooling part **52** may be located upstream of the valve unit **51**.

(64) The wash water that is supplied from the water supply source WS to the cold water flow channel **50** passes through the valve unit **51** and the cooling part **52** and is supplied to the heat exchanger **40**. At this time, the heat transfer plate **80** that absorbs heat from the switching element **60** is water-cooled by the cooling part **52** due to the wash water passing through the cooling part **52**. That is, the heat transfer plate **80** and the cooling part **52** function as a heat sink for dissipating the heat of the switching element **60**.

(65) The heat of the switching element **60** can be effectively dissipated by providing the heat transfer plate **80** for dissipating the heat of the switching element **60** and by providing the cooling part **52** that water-cools the heat transfer plate **80**. Accordingly, the effects of the heat from the switching element **60** on the controller **71** can be suppressed even when the switching element **60** and the controller **71** are mounted to one control board **70**. Because the cold water flow channel **50** includes the cooling part **52**, the wash water from the water supply source WS that is to be supplied to the heat exchanger **40** can be utilized to cool the heat transfer plate **80** before being heated.

Accordingly, it is unnecessary to separately provide a part that cools the heat transfer plate **80**; and the casing **10** can be made smaller. The heat transfer plate **80** (the switching element **60**) can be cooled more efficiently by water-cooling the heat transfer plate **80** than by air-cooling the heat transfer plate **80**.

(66) It is favorable for the heat exchanger **40** to be located distant to the switching element **60**. It is favorable for the cooling part **52** to be located proximate to the switching element **60**. More specifically, it is favorable for a distance L**2** between the heat exchanger **40** and the switching element **60** to be greater than a distance L**1** between the cooling part **52** and the switching element **60**.

(67) By setting the distance L**2** to be greater than the distance L**1**, the heat exchanger **40**, i.e., a heat source, can be located distant to the switching element **60**, and the cooling part **52** can be located proximate to the switching element **60**. The effects of the heat from the heat exchanger **40** on the controller **71** mounted to the same control board **70** as the switching element **60** can be suppressed thereby. Because the heat exchanger **40** can be prevented from being positioned between the cooling part **52** and the switching element **60**, the heating of the water flowing through the cooling part **52** by the heat from the heat exchanger **40** can be suppressed. The heat transfer plate **80** (the switching element **60**) can be more efficiently cooled thereby.

(68) The casing **10** includes a substrate storage part **13** configured to store the control board **70** in the upright state. For example, the control board **70** is located so that the surface on which the circuit is formed faces the longitudinal direction. For example, the control board **70** is located so that the surface to which the switching element **60** is mounted faces frontward. The control board **70** may be located along the vertical direction or may be inclined frontward or backward with respect to the vertical direction. When the control board **70** is inclined, it is favorable for the incline angle to be not more than 45 degrees with respect to the vertical direction, more favorably not more than 30 degrees with respect to the vertical direction, and more favorably not more than 15 degrees with respect to the vertical direction.

(69) For example, the substrate storage part **13** (the control board **70**) is located at the back part of the casing **10**. For example, at least a portion of the substrate storage part **13** is positioned further backward than a longitudinal-direction center CL**1** of the casing **10**. For example, the front end of the substrate storage part **13** is positioned further backward than the longitudinal-direction center CL**1** of the casing **10**.

(70) For example, when the height at the back of the casing **10** is greater than the height at the front of the casing **10**, an enlargement of the casing **10** can be suppressed and the space inside the casing **10** can be effectively used by providing the substrate storage part **13** configured to store the control board **70** in the upright state at the back part of the casing **10**.

(71) For example, the substrate storage part **13** (the control board **70**) is located at one side of the casing **10** in the lateral direction. For example, the toilet lid opening/closing part **26** is located at the other side of the casing **10** in the lateral direction. In the example, the substrate storage part **13** (the control board **70**) is located at the right side of the casing **10**; and the toilet lid opening/closing part **26** is located at the left side of the casing **10**. For example, the substrate storage part **13** (the control board **70**) is located at the side opposite to the toilet lid opening/closing part **26** in the lateral direction with the nozzle **30** interposed.

(72) The space inside the casing **10** can be effectively used by providing the control board **70** at one side of the casing **10** in the lateral direction and by providing the toilet lid opening/closing part **26** at the other side of the casing **10** in the lateral direction. The casing **10** can be made smaller thereby.

(73) The switching element **60** is provided to protrude frontward from the control board **70**. The first switching element **61** and the second switching element **62** are arranged in the lateral direction on the control board **70**.

(74) For example, the switching element **60** is mounted to the lower part of the control board **70**.

For example, at least a portion of the switching element **60** is positioned lower than a vertical-direction center CL**2** of the control board **70**. For example, the upper end of the switching element **60** is positioned lower than the vertical-direction center CL**2** of the control board **70**.
(75) Because it is common for the cold water flow channel **50** to be located at the lower part of the casing **10**, the cooling part **52** can be made smaller by mounting the switching element **60** at the lower part of the control board **70** in the upright state compared to when the switching element **60** is mounted at the upper part of the control board **70** in the upright state. The casing **10** can be made smaller thereby.
(76) For example, the heat transfer plate **80** is located at a position at which the heat transfer plate **80** overlaps the valve unit **51** in the longitudinal direction. For example, the switching element **60** is located at a position at which the switching element **60** overlaps the heat transfer plate **80** in the longitudinal direction. For example, the heat transfer plate **80** is located between the valve unit **51** and the switching element **60** in the longitudinal direction.
(77) By providing the heat transfer plate **80** at a position at which the heat transfer plate **80** overlaps the valve unit **51** in the longitudinal direction and by providing the switching element **60** at a position at which the switching element **60** overlaps the heat transfer plate **80** in the longitudinal direction, the cooling part **52** can be made smaller than when the heat transfer plate **80** is located at a position at which the heat transfer plate **80** does not overlap the valve unit **51** in the longitudinal direction, or when the switching element **60** is located at a position at which the switching element **60** does not overlap the heat transfer plate **80** in the longitudinal direction. The casing **10** can be made smaller thereby.
(78) The flow of the wash water in the cooling part **52** will now be described in more detail.
(79) FIG. **6** is a plan view illustrating the periphery of the switching element of the sanitary washing device according to the embodiment.
(80) FIG. **7** is a cross-sectional view illustrating a portion of the sanitary washing device according to the embodiment.
(81) FIG. **8** is a cross-sectional view illustrating a portion of the sanitary washing device according to the embodiment.
(82) FIG. **9** is a perspective view illustrating a portion of the cold water flow channel of the sanitary washing device according to the embodiment.
(83) The substrate storage part **13** is not illustrated in FIG. **6**.
(84) FIG. **7** is a cross-sectional view along line A**1**-A**2** shown in FIG. **6**.
(85) FIG. **8** is a cross-sectional view along line B**1**-B**2** shown in FIG. **6**.
(86) As illustrated in FIGS. **6** to **9**, the cooling part **52** includes an inflow part **52***a*, an outflow part **52***b*, and first to third flow channels **52***c* to **52***e*.
(87) The inflow part **52***a* is located at the end part of the cooling part **52** at the upstream side. The outflow part **52***b* is located at the end part of the cooling part **52** at the downstream side. In the example, the outflow part **52***b* is connected with the heat exchanger **40**. The first flow channel **52***c*, a second flow channel **52***d*, and the third flow channel **52***e* are located between the inflow part **52***a* and the outflow part **52***b*. The first to third flow channels **52***c* to **52***e* are arranged in the order of the first flow channel **52***c*, the second flow channel **52***d*, and the third flow channel **52***e* from the upstream side.
(88) In the example, the first flow channel **52***c* extends substantially horizontally backward from the inflow part **52***a*. The second flow channel **52***d* detours backward from the connection part between the first flow channel **52***c* and the second flow channel **52***d* and then extends frontward. The connection part between the second flow channel **52***d* and the third flow channel **52***e* is positioned further frontward than the connection part between the first flow channel **52***c* and the second flow channel **52***d*. The third flow channel **52***e* extends upward and frontward from the connection part between the second flow channel **52***d* and the third flow channel **52***e*. The outflow part **52***b* extends substantially horizontally leftward from the upper part of the third flow channel

52*e*.
(89) The water that flows in through the inflow part **52***a* passes through the first flow channel **52***c* and reaches the second flow channel **52***d*. The water that flows through the second flow channel **52***d* passes through the third flow channel **52***e*, reaches the outflow part **52***b*, and flows out to the heat exchanger **40**.
(90) The cooling part **52** contacts the second part **82** of the heat transfer plate **80** at the second flow channel **52***d*. The cooling part **52** water-cools the second part **82** of the heat transfer plate **80** by the water flowing through the second flow channel **52***d*. The cooling efficiency of the heat transfer plate **80** can be increased by providing the second flow channel **52***d* with the structure having the detour.
(91) For example, the heat exchanger **40** is located at a position at which the heat exchanger **40** overlaps the cooling part **52** in the lateral direction. For example, the heat exchanger **40** is located at a position at which the heat exchanger **40** overlaps the inflow part **52***a* in the lateral direction.
(92) FIG. **10** is a perspective view illustrating a portion of the casing of the sanitary washing device according to the embodiment.
(93) Only the case plate **11** is illustrated in FIG. **10**.
(94) As illustrated in FIG. **10**, the case plate **11** includes a water receiving part **14** for receiving condensation water generated at the heat transfer plate **80**.
(95) As illustrated in FIGS. **7** and **8**, at least a portion of the water receiving part **14** is positioned below the heat transfer plate **80** (the second part **82**). The second part **82** of the heat transfer plate **80** is inclined downward from the switching element **60** toward the water receiving part **14**. Accordingly, the condensation water that is generated at the heat transfer plate **80** is discharged into the water receiving part **14** via the second part **82**.
(96) The case plate **11** also includes a drain part **15** for draining the water on the case plate **11** into the bowl of the toilet **200**, and a drain flow channel **16** connecting the water receiving part **14** and the drain part **15**. Broken line arrows in FIG. **10** show the flow of the water when the water is received by the water receiving part **14**, passes through the drain flow channel **16**, and is drained through the drain part **15**.
(97) The drain flow channel **16** guides the water received by the water receiving part **14** to the drain part **15**. For example, the drain flow channel **16** is inclined downward from the water receiving part **14** toward the drain part **15**. In the example as illustrated in FIG. **3**, the drain part **15** is located below the nozzle **30**. The drain part **15** is not limited to such a configuration; it is sufficient for the drain part **15** to be located at a position at which the drain part **15** overlaps the bowl of the toilet **200** in the vertical direction.
(98) The condensation water generated at the heat transfer plate **80** can be prevented from flowing toward the control board **70** side by providing the water receiving part **14** for receiving the condensation water generated at the heat transfer plate **80** below the heat transfer plate **80** and by setting the heat transfer plate **80** to be inclined downward from the switching element **60** toward the water receiving part **14**. Malfunction of the control board **70** due to condensation water can be suppressed thereby.
(99) Embodiments may include the following configurations.
(100) (Configuration 1)
(101) A sanitary washing device, comprising: a nozzle discharging wash water for a private part wash; a heat exchanger using alternating current power to heat the wash water supplied to the nozzle; a cold water flow channel supplying the wash water to the heat exchanger from a water supply source; a switching element switching a supply state of the alternating current power to the heat exchanger; a control board to which a controller is mounted, the controller being operated by direct current power, the controller controlling the switching element; a heat transfer plate configured to dissipate heat of the switching element; and a casing storing the nozzle, the heat exchanger, the cold water flow channel, the switching element, the control board, and the heat

transfer plate, the switching element being mounted to the control board, the cold water flow channel including a cooling part, the cooling part water-cooling the heat transfer plate.
(Configuration 2)
(102) The device according to configuration 1, wherein the heat exchanger is an instantaneous heat exchanger, and a distance between the heat exchanger and the switching element is greater than a distance between the cooling part and the switching element.
(Configuration 3)
(103) The device according to configuration 1 or 2, wherein the casing includes a substrate storage part configured to store the control board in an upright state, the substrate storage part is located at a back part of the casing; and the switching element is mounted to a lower part of the control board.
(Configuration 4)
(104) The device according to any one of configurations 1 to 3, further comprising: a valve unit located in the cold water flow channel, the heat transfer plate being located at a position at which the heat transfer plate overlaps the valve unit in a longitudinal direction, the switching element being located at a position at which the switching element overlaps the heat transfer plate in the longitudinal direction.
(Configuration 5)
(105) The device according to any one of configurations 1 to 4, wherein the casing includes a water receiving part, at least a portion of the water receiving part being positioned below the heat transfer plate, the water receiving part being configured to receive condensation water generated at the heat transfer plate, the heat transfer plate being inclined downward from the switching element toward the water receiving part.
(Configuration 6)
(106) The device according to any one of configurations 1 to 5, further comprising: a toilet lid; and a toilet lid opening/closing part configured to open and close the toilet lid, the control board being located at one side of the casing in a lateral direction, the toilet lid opening/closing part being located at an other side of the casing in the lateral direction.
(107) Thus, according to embodiments, a sanitary washing device is provided in which the casing can be made smaller, and the effects of the heat from the switching element on the controller can be suppressed even when the switching element and the controller are mounted to one control board.
(108) The invention has been described with reference to the embodiments. However, the invention is not limited to these embodiments. Any design changes in the above embodiments suitably made by those skilled in the art are also encompassed within the scope of the invention as long as they fall within the spirit of the invention. For example, the shape, the size the material, the disposition and the arrangement or the like of the components included in the sanitary washing device are not limited to illustrations and can be changed appropriately.
(109) The components included in the embodiments described above can be combined to the extent possible, and these combinations are also encompassed within the scope of the invention as long as they include the features of the invention.

Claims

1. A sanitary washing device, comprising: a nozzle discharging wash water for a private part wash; a heat exchanger using alternating current power to heat the wash water supplied to the nozzle; a cold water flow channel supplying the wash water to the heat exchanger from a water supply source; a switching element switching a supply state of the alternating current power to the heat exchanger; a control board to which a controller is mounted, the controller being operated by direct current power, the controller controlling the switching element; a heat transfer plate configured to dissipate heat of the switching element; a casing storing the nozzle, the heat exchanger, the cold water flow channel, the switching element, the control board, and the heat transfer plate; and a

valve unit located in the cold water flow channel, the switching element being mounted to the control board, the cold water flow channel including a cooling part, the cooling part water-cooling the heat transfer plate, the casing including a substrate storage part configured to store the control board in an upright state, the substrate storage part being located at a back part of the casing, the switching element being mounted to a lower part of the control board, the heat transfer plate being located at a position at which the heat transfer plate overlaps the valve unit in a longitudinal direction, the switching element being located at a position at which the switching element overlaps the heat transfer plate in the longitudinal direction.

2. The device according to claim 1, wherein the heat exchanger is an instantaneous heat exchanger, and a distance between the heat exchanger and the switching element is greater than a distance between the cooling part and the switching element.

3. The device according to claim 1, wherein the casing includes a water receiving part, at least a portion of the water receiving part being positioned below the heat transfer plate, the water receiving part being configured to receive condensation water generated at the heat transfer plate, the heat transfer plate being inclined downward from the switching element toward the water receiving part.

4. The device according to claim 1, further comprising: a toilet lid; and a toilet lid opening/closing part configured to open and close the toilet lid, the control board being located at one side of the casing in a lateral direction, the toilet lid opening/closing part being located at an other side of the casing in the lateral direction.

5. A sanitary washing device, comprising: a nozzle discharging wash water for a private part wash; a heat exchanger using alternating current power to heat the wash water supplied to the nozzle; a cold water flow channel supplying the wash water to the heat exchanger from a water supply source; a switching element switching a supply state of the alternating current power to the heat exchanger; a control board to which a controller is mounted, the controller being operated by direct current power, the controller controlling the switching element; a heat transfer plate configured to dissipate heat of the switching element; and a casing storing the nozzle, the heat exchanger, the cold water flow channel, the switching element, the control board, and the heat transfer plate, the switching element being mounted to the control board, the cold water flow channel including a cooling part, the cooling part water-cooling the heat transfer plate, the casing including a substrate storage part configured to store the control board in an upright state, the substrate storage part being located at a back part of the casing, the switching element being mounted to a lower part of the control board, the casing including a water receiving part, at least a portion of the water receiving part being positioned below the heat transfer plate, the water receiving part being configured to receive condensation water generated at the heat transfer plate, the heat transfer plate being inclined downward from the switching element toward the water receiving part.
